

Q1 - 24 June - Shift 1

Let $S = \{\sqrt{n} : 1 \leq n \leq 50 \text{ and } n \text{ is odd}\}$

Space for your notes:

Let $a \in S$ and $A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$

If $\sum_{a \in S} \det(\text{adj} A) = 100\lambda$, then λ is equal to

(A) 218

(B) 221

(C) 663

(D) 1717

Q2 - 24 June - Shift 2

Let $S = \left\{ \begin{pmatrix} -1 & a \\ 0 & b \end{pmatrix} ; a, b \in \{1, 2, 3, \dots, 100\} \right\}$ and let

Space for your notes:

$T_n = \{A \in S : A^{n(n+1)} = I\}$. Then the number of

elements in $\bigcap_{n=1}^{100} T_n$ is _____.

Q3 - 25 June - Shift 1

Let A be a 3×3 real matrix such that

$$A \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}; A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \text{ and } A \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

If $X = (x_1, x_2, x_3)^T$ and I is an identity matrix

of order 3, then the system $(A - 2I)X = \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}$

has

- (A) no solution
- (B) infinitely many solutions
- (C) unique solution
- (D) exactly two solutions

Q4 - 25 June - Shift 1

Let $A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. If M and N are two matrices

given by $M = \sum_{k=1}^{10} A^{2k}$ and $N = \sum_{k=1}^{10} A^{2k-1}$ then MN^2 is

- (A) a non-identity symmetric matrix
- (B) a skew-symmetric matrix
- (C) neither symmetric nor skew-symmetric matrix
- (D) an identity matrix

Q5 - 25 June - Shift 1

Let A be a 3×3 matrix having entries from the set $\{-1, 0, 1\}$. The number of all such matrices A having sum of all the entries equal to 5, is _____

Space for your notes:

Space for your notes:

Space for your notes:

Q6 - 25 June - Shift 2

Let $A = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$. Then the

number of elements in the set

$\{(n, m) : n, m \in \{1, 2, \dots, 10\} \text{ and } nA^n + mB^m = I\}$

is _____

Space for your notes:

Q7 - 26 June - Shift 1

Let A be a 3×3 invertible matrix. If $|\text{adj}(24A)| = |\text{adj}(3\text{adj}(2A))|$, then $|A|^2$ is equal to :

(A) 6^6 (B) 2^{12} (C) 2^6 (D) 1

Space for your notes:

Q8 - 26 June - Shift 2

Let $X = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $Y = \alpha I + \beta X + \gamma X^2$ and

$Z = \alpha^2 I - \alpha\beta X + (\beta^2 - \alpha\gamma)X^2$, $\alpha, \beta, \gamma \in \mathbb{R}$. If $Y^{-1} =$

$\begin{bmatrix} \frac{1}{5} & \frac{-2}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & \frac{-2}{5} \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$, then $(\alpha - \beta + \gamma)^2$ is equal to _____.

Space for your notes:

Q9 - 27 June - Shift 1

The positive value of the determinant of the matrix

$$A, \text{ whose } \text{Adj}(\text{Adj}(A)) = \begin{pmatrix} 14 & 28 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{pmatrix},$$

is _____.

Space for your notes:

Q10 - 27 June - Shift 2

Let A and B be two 3×3 matrices such that

$$AB = I \text{ and } |A| = \frac{1}{8} \text{ then } |\text{adj}(B \text{adj}(2A))| \text{ is equal to}$$

- (A) 16 (B) 32
(C) 64 (D) 128

Space for your notes:

Q11 - 27 June - Shift 2

Let A be a matrix of order 2×2 , whose entries are from the set $\{0, 1, 2, 3, 4, 5\}$. If the sum of all the entries of A is a prime number p, $2 < p < 8$, then the number of such matrices A is :

Space for your notes:

Q12 - 28 June - Shift 1

Let A be a matrix of order 3×3 and $\det(A) = 2$.

Then $\det(\det(A) \text{adj}(5 \text{adj}(A^3)))$ is equal to _____.

- (A) 512×10^6 (B) 256×10^6
(C) 1024×10^6 (D) 256×10^{11}

Space for your notes:

Q13 - 28 June - Shift 2

Let $A = \begin{pmatrix} 1+i & 1 \\ -i & 0 \end{pmatrix}$ where $i = \sqrt{-1}$.

Space for your notes:

Then, the number of elements in the set

$\{n \in \{1, 2, \dots, 100\} : A^n = A\}$ is

Q14 - 29 June - Shift 1

The probability that a randomly chosen 2×2 matrix with all the entries from the set of first 10 primes, is singular, is equal to :

Space for your notes:

- (A) $\frac{133}{10^4}$ (B) $\frac{18}{10^3}$
 (C) $\frac{19}{10^3}$ (D) $\frac{271}{10^4}$

Q15 - 29 June - Shift 1

Let $A = [a_{ij}]$ be a square matrix of order 3 such that $a_{ij} = 2^{j-i}$, for all $i, j = 1, 2, 3$. Then, the matrix $A^2 + A^3 + \dots + A^{10}$ is equal to :

Space for your notes:

- (A) $\left(\frac{3^{10}-3}{2}\right)A$ (B) $\left(\frac{3^{10}-1}{2}\right)A$
 (C) $\left(\frac{3^{10}+1}{2}\right)A$ (D) $\left(\frac{3^{10}+3}{2}\right)A$

Q16 - 29 June - Shift 2

Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$. If $B = I - {}^5C_1 (\text{adj}A) + {}^5C_2 (\text{adj}A)^2 - \dots - {}^5C_5 (\text{adj}A)^5$, then the sum of all elements of the matrix B is:

- (A) -5 (B) -6
(C) -7 (D) -8

Space for your notes:

Q17 - 29 June - Shift 2

Let $M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$, where α is a non-zero real number and $N = \sum_{k=1}^{49} M^{2k}$. If $(I - M^2)N = -2I$, then the positive integral value of α is _____.

Space for your notes:

Answer Key**Q1 (B)****Q2 (100)****Q3 (B)****Q4 (A)****Q5 (414)****Q6 (1)****Q7 (C)****Q8 (100)****Q9 (14)****Q10 (C)****Q11 (180)****Q12 (A)****Q13 (25)****Q14 (C)****Q15 (A)****Q16 (C)****Q17 (1)**

#MathBoleTohMathonGo

Q1 (B)

$$S = \{\sqrt{n} : 1 \leq n \leq 50 \text{ and } n \text{ is odd}\}$$

$$= \{\sqrt{1}, \sqrt{3}, \sqrt{5}, \dots, \sqrt{49}\}, 25 \text{ terms}$$

$$|A| = 1 + a^2$$

$$\sum_{a \in S} \det(\text{adj} A) = \sum_{a \in S} |A|^2 = \sum (1 + a^2)^2$$

$$= 22100 = 100\lambda$$

$$\lambda = 221$$

Q2 (100)

$$A = \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix} \begin{bmatrix} -1 & a \\ 0 & b \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -a + ab \\ 0 & b^2 \end{bmatrix}$$

$$\therefore T_n = \{A \in S; A^{n(n+1)} = I\}$$

$\therefore b$ must be equal to 1

$\therefore$ In this case A^2 will become identity matrix and a can take any value from 1 to 100

$\therefore$ Total number of common element will be 100.

Q3 (B)

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$A \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow c_1 = 1, c_2 = 1, c_3 = 2$$

$$A \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 + a_1 \\ c_2 + a_2 \\ c_3 + a_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow a_1 = -2, a_2 = -1, a_3 = -1$$

$$A \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow b_1 = 3, b_2 = 2, b_3 = 1$$

$$\Rightarrow A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

$$\Rightarrow A - 2I = \begin{bmatrix} -4 & 3 & 1 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$|A - 2I| = 0$$

$$\text{Now, } \begin{bmatrix} -4 & 3 & 1 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$$

$$-4x_1 + 3x_2 + x_3 = 4 \quad \dots(1)$$

$$-x_1 + x_3 = 1 \quad \dots(2)$$

$$-x_1 + x_2 = 1 \quad \dots(3)$$

$$(1) - [(2) + 3(3)]$$

$$0 = 0 \Rightarrow \text{infinite solutions}$$

#MathBoleTohMathonGo

To practice more chapter-wise JEE Main PYQs, [click here to download the MARKS app](#) from Playstore

Q4 (A)

$$A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix} = -4I$$

$$A^3 = -4A$$

$$A^4 = (-4I)(-4I) = (-4)^2 I$$

$$A^5 = (-4)^2 A, \quad A^6 = (-4)^3 I$$

$$M = \sum_{k=1}^{10} A^{2k} = A^2 + A^4 + \dots + A^{20}$$

$$= [-4 + (-4)^2 + (-4)^3 + \dots + (-4)^{10}]I$$

$$= -4I$$

$\Rightarrow$ M is symmetric matrix

$$N = \sum_{k=1}^{10} A^{2k-1} = A + A^3 + \dots + A^{19}$$

$$= A[1 + (-4) + (-4)^2 + \dots + (-4)^9]$$

$$= \lambda A \Rightarrow \text{skew symmetric}$$

$\Rightarrow N^2$ is symmetric matrix

$\Rightarrow MN^2$ is non identity symmetric matrix

Q5 (414)

Case-I: 1 $\rightarrow$ 7 times

and - 1 $\rightarrow$ 2 times

$$\text{number of possible matrix} = \frac{9!}{7! 2!} = 36$$

Case-II: 1 $\rightarrow$ 6 times,

- 1 $\rightarrow$ 1 times

and 0 $\rightarrow$ 2 times

$$\text{number of possible matrix} = \frac{9!}{6! 2!} = 252$$

Case-III: 1 $\rightarrow$ 5 times,

and 0 $\rightarrow$ 4 times

$$\text{number of possible matrix} = \frac{9!}{5! 4!} = 126$$

Hence total number of all such matrix A

$$= 414$$

Q6 (I)

$$A^2 = A \text{ and } B^2 = B$$

Therefore equation $nA^n + mB^m = I$ becomes

$$nA + mB = I, \text{ which gives } m = n = 1$$

Only one set possible

Q7 (C)

$$|\text{adj}(24A)| = |\text{adj } 3(\text{adj } 2A)|$$

$$\Rightarrow |24A|^2 = (3 |\text{adj}(2A)|)^2$$

$$\Rightarrow (24^3 |A|)^2 = (3^3 |\text{adj}(2A)|)^2$$

$$= 3^6 (|2A|)^2$$

$$\Rightarrow 24^6 |A|^2 = (24^3 |A|)^2 = 3^6 \times 2^{12} |A|^4$$

$$\Rightarrow |A|^2 = \frac{24^6}{3^6 \times 2^{12}} = 64$$

Q8 (100)

$$X = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, X^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} \alpha & \beta & \gamma \\ 0 & \alpha & \beta \\ 0 & 0 & \alpha \end{bmatrix}, Z = \begin{bmatrix} \alpha^2 & -\alpha\beta & \beta^2 - \alpha\gamma \\ 0 & \alpha^2 & -\alpha\beta \\ 0 & 0 & \alpha^2 \end{bmatrix}$$

$$Y \cdot Y^{-1} = I$$

$$\begin{bmatrix} \alpha & \beta & \gamma \\ 0 & \alpha & \beta \\ 0 & 0 & \alpha \end{bmatrix} \begin{bmatrix} \frac{1}{\alpha} & -\frac{\beta}{\alpha^2} & \frac{\gamma}{\alpha^2} \\ 0 & \frac{1}{\alpha} & -\frac{\beta}{\alpha^2} \\ 0 & 0 & \frac{1}{\alpha} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\frac{\alpha}{\alpha} = 1 \Rightarrow \alpha = 5$$

$$-\frac{2}{5}\alpha + \frac{\beta}{5} = 0 \Rightarrow \beta = 10$$

$$\frac{\alpha}{5} - \frac{2\beta}{5} + \frac{\gamma}{5} = 0 \Rightarrow \gamma = 15$$

$$\Rightarrow (\alpha - \beta + \gamma)^2 = (5 - 10 + 15)^2 = 100$$

Q9 (14)

#MathBoleTohMathonGo

$$\text{Adj}(\text{Adj}A) = \begin{bmatrix} 14 & 18 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{bmatrix}$$

$$|\text{Adj}(\text{Adj}A)| = \begin{vmatrix} 14 & 28 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{vmatrix} = 14 \times 14 \times 14 \begin{vmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{vmatrix}$$

$$= (14)^3 [3 - 2(-5) - 1(-1)] = (14)^3 [14] = (14)^4$$

$$|A|^4 = (14)^4 \Rightarrow |A| = 14$$

Q10 (C)

$$AB = i$$

$$|\text{adj}(B \text{ adj}(2A))| = |B \text{ adj}(2A)|^2$$

$$= |B|^2 |\text{adj}(2A)|^2$$

$$= |B|^2 (|2A|^2)^2 = |B|^2 (2^6 |A|^2)^2$$

$$|A| = \frac{1}{8} \text{ and } |AB| = 1 \Rightarrow |A| |B| = 1$$

$$\Rightarrow \frac{1}{8} |B| = 1$$

$$\Rightarrow |B| = 8$$

$$\text{required value} = 64$$

Q11 (180)

Hints and Solutions

MathonGo

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$; $a, b, c, d \in \{0, 1, 2, 3, 4, 5\}$

$$a + b + c + d = p, p \in \{3, 5, 7\}$$

Case-(i)

$$a + b + c + d = 3; a, b, c, d \in \{0, 1, 2, 3\}$$

$$\text{No. of ways} = {}^{3+4-1}C_{4-1} = {}^6C_3 = 56 \quad \dots\dots (1)$$

Case-(ii)

$$a + b + c + d = 5; a, b, c, d \in \{0, 1, 2, 3, 4, 5\}$$

$$\text{No. of ways} = {}^{5+4-1}C_{4-1} = {}^8C_3 = 56 \quad \dots\dots (2)$$

Case-(iii)

$$a + b + c + d = 7$$

$$\text{No. of ways} = \text{total ways when } a, b, c, d \in \{0, 1, 2,$$

$$3, 4, 5, 6, 7\} - \text{total ways when } a, b, c, d \notin \{6, 7\}$$

$$\text{No of ways} = {}^{7+4-1}C_{4-1} = \left(\frac{4}{3} + \frac{4}{2} \right)$$

$$= {}^{10}C_3 - 16 = 104 \quad \dots\dots (3)$$

$$\text{Hence total no. of ways} = 180$$

Q12 (A)

$$|(\det(A)) \operatorname{adj}(5 \operatorname{adj}(A))|$$

$$= |2 \operatorname{adj}(5 \operatorname{adj}(A^3))|$$

$$= 2^3 |\operatorname{adj}(5 \operatorname{adj}(A^3))|$$

$$= 2^3 \cdot |5 \operatorname{adj}(A^3)|^2$$

$$= 2^3 (5^3 \cdot |\operatorname{adj}(A^3)|)^2$$

$$= 2^3 \cdot 5^6 \cdot |\operatorname{adj} A^3|^2$$

$$= 2^3 \cdot 5^6 ((|A|^3)^2)^2$$

$$= 2^3 \cdot 5^6 \cdot 2^{12} = 2^{15} \times 5^6$$

$$= 2^9 \times 10^6$$

$$= 512 \times 10^6.$$

Q13 (25)

$$A = \begin{bmatrix} 1+i & 1 \\ -i & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1+i & 1 \\ -i & 0 \end{bmatrix} \begin{bmatrix} 1+i & 1 \\ -i & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} i & 1+i \\ -i+1 & -i \end{bmatrix}$$

$$A^4 = \begin{bmatrix} i & 1+i \\ -i+1 & -i \end{bmatrix} \begin{bmatrix} i & 1+i \\ -i+1 & -i \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$A^{4n+1} = A$$

$$n = 1, 5, 9, \dots, 97$$

$\Rightarrow$ total elements in the set is 25.

Q14 (C)

Let matrix A is singular then $|A| = 0$

Number of singular matrix = All entries are same +
only two prime number are used in matrix

$$= 10 + 10 \times 9 \times 2$$

$$= 190$$

$$\text{Required probability} = \frac{190}{10^4} = \frac{19}{10^3}$$

Q15 (A)

$$A = \begin{pmatrix} 1 & 2 & 2^2 \\ 1/2 & 1 & 2 \\ 1/2^2 & 1/2 & 1 \end{pmatrix}$$

$$A^2 = 3A$$

$$A^3 = 3^2 A$$

$$A^2 + A^3 + \dots + A^{10}$$

$$= 3A + 3^2 A + \dots + 3^9 A = \frac{3(3^9 - 1)}{3 - 1} A$$

$$= \frac{3^{10} - 3}{2} A$$

Q16 (C)

$$B = (I - \text{adj}A)^5 = \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}^5 = \begin{bmatrix} -1 & -5 \\ 0 & -1 \end{bmatrix}$$

Sum of its all elements = -7.

Q17 (I)

$$M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}; M^2 = \begin{bmatrix} -\alpha^2 & 0 \\ 0 & -\alpha^2 \end{bmatrix} = -\alpha^2 I$$

$$N = M^2 + M^4 + \dots + M^{98} = [-\alpha^2 + \alpha^4 - \alpha^6 + \dots] I$$

$$= -\alpha^2 \frac{(1 - (-\alpha^2)^{49})}{1 + \alpha^2} \cdot I$$

$$I - M^2 = (1 + \alpha^2) I$$

$$(I - M^2)N = -\alpha^2 (\alpha^{98} + 1) = -2$$

$$\alpha = 1$$